Module 06 Lab - Regression and Classification Models**Objective:** To understand the difference between regression and classification and to build your first linear models for both tasks.**In this lab, you will write the code to train and evaluate the models.**

Part 1: Regression vs. ClassificationThis is the most fundamental distinction between types of supervised learning problems.* **Regression:** The goal is to predict a **continuous numerical value**. * *Examples:* Predicting the price of a house, the temperature tomorrow, or the stock price.* **Classification:** The goal is to predict a **discrete category or class label**. * *Examples:* Predicting if an email is spam or not spam, if a flower is a setosa, versicolor, or virginica, or if a customer will churn or not.**In this lab, we will tackle one of each.**

Part 2: Linear Regression**Concept:** Linear Regression is used to predict a continuous value. It works by finding the best-fitting straight line through the data points. The model learns a "slope" (coefficient) for each feature and an "intercept".* **Problem:** We will predict the `Fare` of a Titanic passenger based on their `Age` and `Pclass`.

```
import pandas as pdfrom sklearn.model_selection import train_test_splitfrom sklearn.linear_model import LinearRegressionfrom
```

```
  File "/tmp/ipykernel_3774/4171891673.py", line 1
    import pandas as pdfrom sklearn.model_selection import train_test_splitfrom sklearn.linear_model import
LinearRegressionfrom sklearn.metrics import mean_squared_error# Load and prepare the datadf =
pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')# For simplicity, we'll drop
rows with missing agedf.dropna(subset=['Age'], inplace=True)# Define features and targetfeatures = ['Age',
'Pclass']target_reg = 'Fare'X_reg = df[features]y_reg = df[target_reg]# Split the dataX_train_reg, X_test_reg, y_train_reg,
y_test_reg = train_test_split(X_reg, y_reg, test_size=0.2, random_state=42)
                          ^
SyntaxError: invalid syntax
```

Next steps: Explain error

Task 1: Train and Evaluate a Linear Regression Model**Your Task:**1. Create an instance of the `LinearRegression` model.2. Train the model using the training data (`X_train_reg`, `y_train_reg`).3. Make predictions on the test data.4. Evaluate the model using `mean_squared_error`. This metric tells us the average of the squared differences between the predicted and actual values.

```
# --- ENTER YOUR CODE HERE ---# 1. Create the model instance# lr_model = ...# 2. Train the model# lr_model.fit(...)# 3. Mak
```

Part 3: Logistic Regression**Concept:** Despite its name, Logistic Regression is a **classification** algorithm. It works by calculating the probability that a given input belongs to a certain class. It's one of the most widely used and interpretable classification models.* **Problem:** We will predict whether a passenger `Survived` based on their `Age`, `Pclass`, and `Sex`.

```
from sklearn.linear_model import LogisticRegressionfrom sklearn.metrics import accuracy_score# Prepare data for classificat
```

Task 2: Train and Evaluate a Logistic Regression Model**Your Task:**1. Create an instance of the `LogisticRegression` model.2. Train the model using the classification training data.3. Make predictions on the test data.4. Evaluate the model using `accuracy_score`.

```
    # --- ENTER YOUR CODE HERE ---# 1. Create the model instance# log_model = ...# 2. Train the model# log_model.fit(...)# 3. M
```

📝 Knowledge Check**Instructions:** Answer the following questions in this markdown cell.1. **In your own words, what is the key difference between a regression problem and a classification problem?**2. **The `LinearRegression` model has an attribute called `.coef_`. After you train the model, print `lr_model.coef_`. What do these numbers represent?3. Why did we use `mean_squared_error` to evaluate the regression model but `accuracy_score` for the classification model? Why wouldn't accuracy be a good metric for the fare prediction task?[ENTER YOUR ANSWERS HERE]**